

■ 2.6.21 Special Topic: Calling External Functions

Functions like `RunThrough` allow you to send input to an external program, and get back the results it generates. In many cases, you may not want to send input to the *whole* external program, but rather, you may want to call just one of the functions in the program.

On most computer systems, *Mathematica* provides a mechanism for doing this. The basic idea is for the external program to accept “packets” of input, that specify which function is to be called, and with what arguments.

There is a collection of utility functions, distributed in C, FORTRAN and other source code forms, which allow you to set up routines in the external program which “decode” packets received from *Mathematica*, and call the necessary functions.

There are also utility functions which allow the external program to send packets back to *Mathematica*, to request results from *Mathematica*.

These functions are discussed in slightly more detail in Section ??.

<code>StartProcess["command"]</code>	start an external program, and set up <i>Mathematica</i> functions to call the functions inside it
<code>CallProcess["command", f, arg₁, arg₂, ...]</code>	call the function <i>f</i> in the external program
<code>EndProcess["command"]</code>	terminate the external program

Calling functions in external programs.

When you call the function `StartProcess`, it typically sets up a sequence of definitions for *Mathematica* functions which call functions in the external program, using `CallProcess`.